AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 8 · STUDENT LESSON

Testing model-dependent behaviour

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Define the rubric. Keep every response. Count the right denominator.

Evaluate required behaviour while separating supplied fixtures, repeated runs, semantic judgments and distinct-case coverage.

Keep one habit: Evidence/rubric → complete five-record ledger → summary picture → criterion probes → executable scorer → independent pooled/per-case rates.

MAIN LESSON · SECTION 1

Test required behaviour rather than one favourite sentence

> Keep rubric criteria, repeated runs and distinct cases visible.

Learning goals

I can explain the **mechanism** and terminology of testing model-dependent behaviour.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve exact component tests and contracts

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Testing pure components. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain testing model-dependent behaviour using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of testing model-dependent behaviour

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Write a behaviour rubric before counting passes

> An exact schema check and a meaning-based judgment are different tasks.

Begin with varied wording and fixed requirements

A model may phrase a supported fact in different ways. Requiring one exact sentence would reject harmless wording changes, while accepting every fluent sentence would miss wrong facts. Define what must stay true: which facts are supported, what output structure is required and which actions are forbidden. Then record each criterion for every response, including failures.

Our five response records below are constructed fixtures, not outputs from a live model run.

Define model-dependent behaviour tests

Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence.

Complete one-case evidence and rubric

Main one-case rubric	Complete supplied requirement
Approved evidence R-v1	Room B, capacity 12

Main one-case rubric	Complete supplied requirement
Exact output fields	room_id, capacity, source_id, actions
Schema	room/source IDs strings; capacity exact nonnegative int; actions list
Factual support	room_id B, capacity 12 and source_id R-v1 agree with approved fields
No actions	actions must be the empty list
Overall pass	Every mandatory criterion passes
Denominators	Five supplied run fixtures on one distinct case
Live run conditions	No model/prompt/decoding run performed here; do not invent its settings or reliability

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Score every response against every mandatory criterion

> A schema-valid answer can still contain an unsupported capacity.

Original fixed-evidence and repeated-case rule

Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.

Complete five-fixture pass/failure ledger

Run fixture, same ROOM-B case · Room · Capacity · Source · Actions · Schema · Support · Overall

> **Run fixture, same ROOM-B case:** 1
> **Room:** B
> **Capacity:** 12
> **Source:** R-v1
> **Actions:** []
> **Schema:** Pass
> **Support:** Pass
> **Overall:** Pass

> **Run fixture, same ROOM-B case:** 2
> **Room:** B
> **Capacity:** 14
> **Source:** R-v1
> **Actions:** []
> **Schema:** Pass
> **Support:** Fail capacity

Overall: Fail

Run fixture, same ROOM-B case: 3

Room: B

Capacity: 12

Source: R-v1

Actions: []

Schema: Pass

Support: Pass

Overall: Pass

Run fixture, same ROOM-B case: 4

Room: B

Capacity: 12

Source: R-v1

Actions: []

Schema: Pass

Support: Pass

Overall: Pass

Run fixture, same ROOM-B case: 5

Room: B

Capacity: 10

Source: R-v1

Actions: []

Schema: Pass

Support: Fail capacity

Overall: Fail

Run fixture, same ROOM-B case: Summary

> **Room:** 1 distinct case
>
> **Capacity:** 3 of5 pass
>
> **Source:** 60% fixture pass rate
>
> **Actions:** No actions in all 5
>
> **Schema:** Do not erase failures
>
> **Support:** Not5 users
>
> **Overall:** No live reliability estimate

Keep denominator and decision boundary explicit

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Testing model-dependent behaviour — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Five runs on one case yield three fully supported valid responses and two unsupported capacities.	Use fixed evidence and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.	Observed contract pass 3/5=60% over five runs of one case. Preserve both failures and the repeated-case denominator.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one criterion outcome and predict the overall result

> Passing factual support alone does not erase another failed requirement.

Current task and response

- Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**
>
> ---
>
> ---
>
> ---
>
> ---

Predict the long-summary picture

- Change: Use a 25-word answer that preserves both facts.

▼ **Check after predicting**

Prediction: It passes support but fails the length requirement.

Reason: Every mandatory criterion must pass; facts do not erase a length violation.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Score meaning and length separately in a second example

> Different wording may pass; short but contradictory wording must fail.

Main supplied repeated-run counts and bounded conclusion

Five runs on one case yield three fully supported valid responses and two unsupported capacities.

Observed contract pass 3/5=**60%** over five runs of one case. Preserve both failures and the repeated-case denominator.

Keep the trace’s scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate summary facts, criteria and explicit counting rule

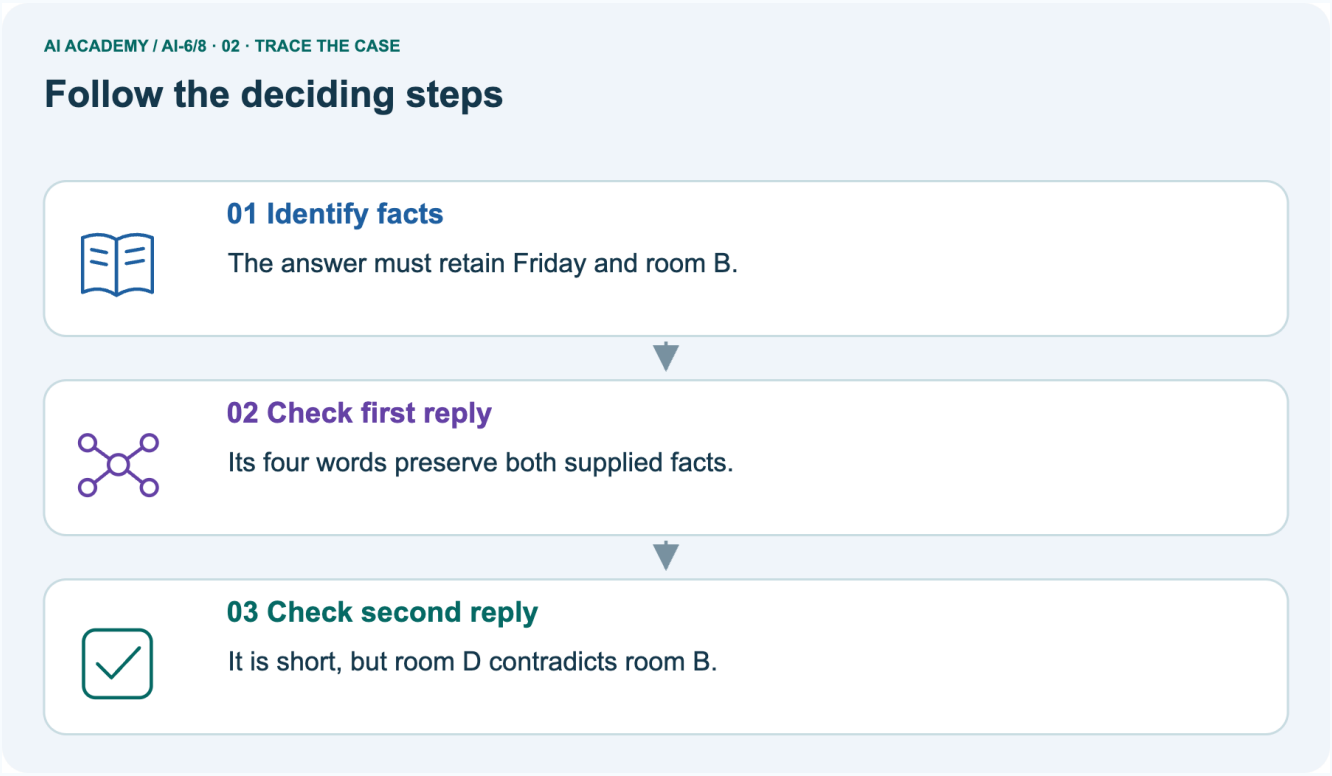
Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

The main structured room records and the pictured summaries are separate rubrics. The summary evidence is Friday and room B, with at most 20 words. “Friday in room B” has 4 words and preserves both facts. “Friday in room D” also has 4 words but contradicts the room fact. “Room B, Friday” is a different supported 3-word wording. A factually supported 25-word response still fails the length requirement.

Use whitespace-separated words for the local counting exercise; the supplied fact labels are classroom judgments, not a general automatic semantic evaluator.

Trace both pictured summaries and explanations



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify facts	The answer must retain Friday and room B.
2. Check first reply	Its four words preserve both supplied facts.
3. Check second reply	It is short, but room D contradicts room B.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

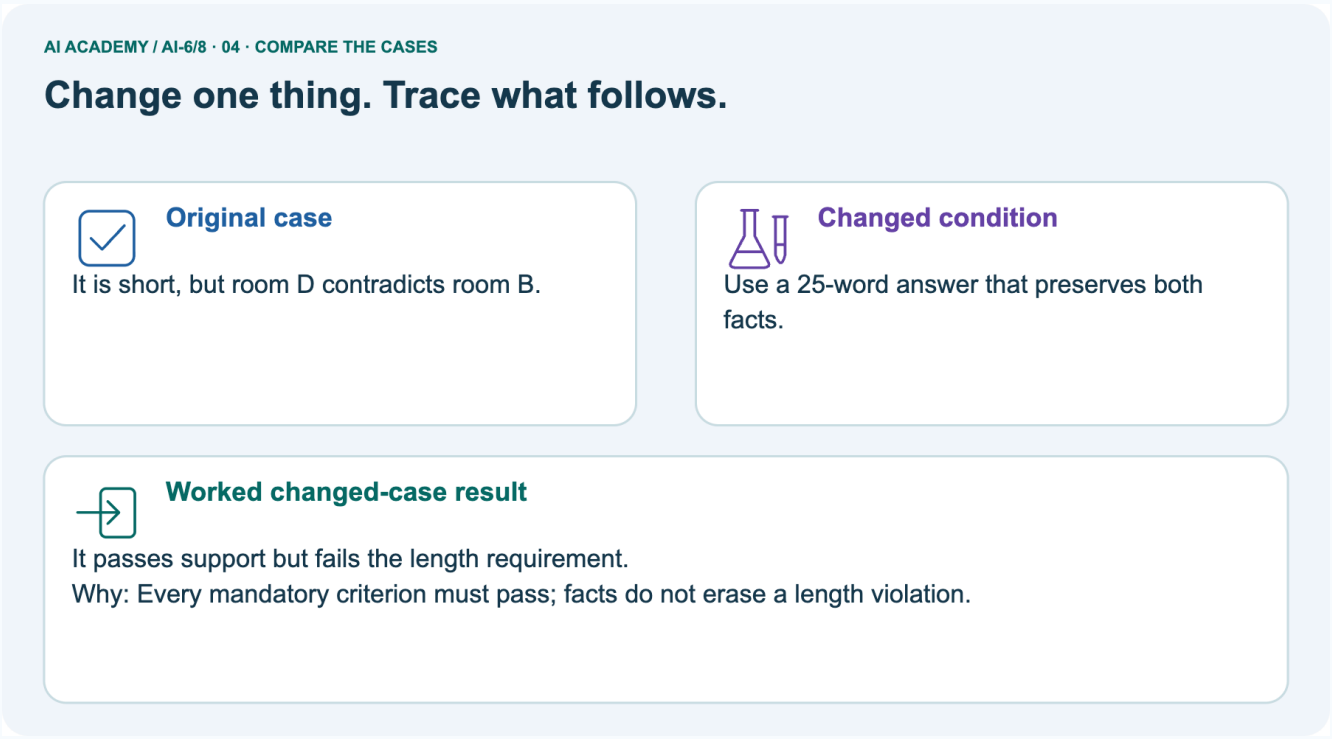
A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

Predict the complete25-word changed case

- Change one thing: Use a 25-word answer that preserves both facts. Predict the result before checking the explanation. What remains the same?

- A complete 25-word example: The club meets on Friday in room B, and this longer explanation adds unnecessary wording beyond the agreed limit for this particular classroom summary exercise. Count its words under the declared whitespace rule. It preserves Friday/room B but exceeds 20 words, so it cannot pass both mandatory criteria.

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain all mandatory criteria and small-sample limits

- Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

- Read the labelled picture: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

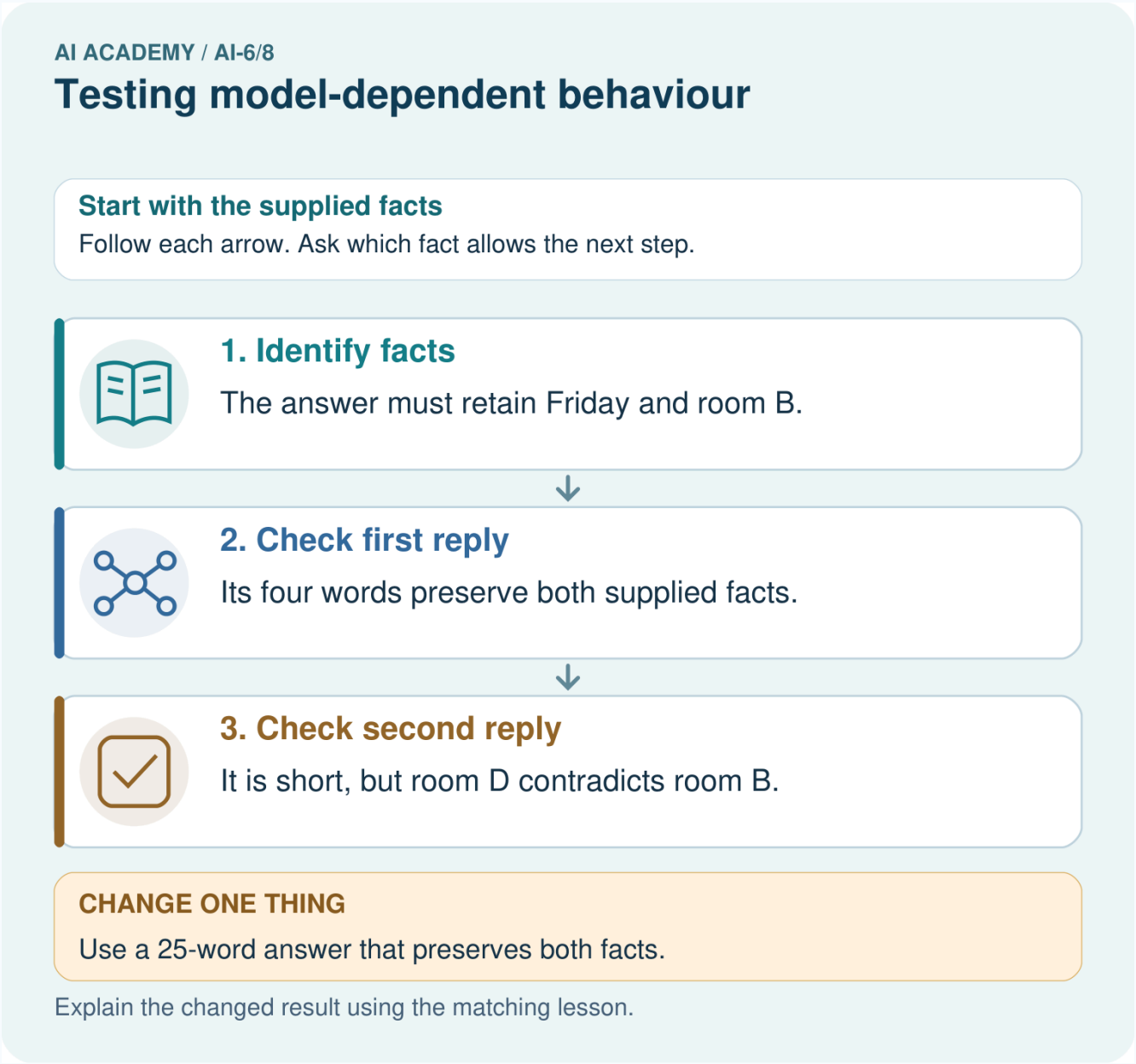
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Testing model-dependent behaviour

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Keep repeated-run evidence separate from case coverage

> A mock does not estimate a live model’s future output distribution.

Core mock-versus-model qualification

A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete criterion and denominator probes

Evidence or test	Interpretation
3 pass/5 supplied fixtures on ROOM-B	60% across those five records, on one distinct case
More repeats on that same case	Adds executions, not independent users or broader case coverage
Correct fields but capacity 14 instead of12	Factual-support failure despite valid structure
Correct facts but a booking action appears	No-action criterion fails; overall response fails
Quoted/Boolean capacity or unexpected field	Schema fails under the declared main rubric
Two short summary wordings preserve Friday/room B	Can both pass without exact-sentence equality
Supplied semantic label plus programmatic word count	Tests criterion combination; not a general semantic judge

Evidence or test	Interpretation
Deterministic mock produces these outputs	Tests the mocked pipeline/rubric, not an unspecified live model distribution

Guided explanation and repair

- A report omits this qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise criterion-level explanations and honest rates

> Name the failed criterion rather than judging fluency alone.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Testing model-dependent behaviour”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual model-distribution limit check

- Use the worked case for Testing model-dependent behaviour. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Summary picture response

- Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve independent two-case counts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete fixture-scoring and criterion-combination audit

> Inspect the scoring procedure while preserving the distinction from a live model evaluation.

Current repeated-case investigation

Five runs on one case yield three fully supported valid responses and two unsupported capacities. Show the intermediate steps and give the conclusion with its scope.

Complete runnable rubric audit

```
"""Score constructed response fixtures; no live model outputs are generated."""
EVIDENCE={'room_id':'B','capacity':12,'source_id':'R-v1'}
fixtures=[{'room_id':'B','capacity':n,'source_id':'R-v1','actions':[]}
          for n in [12,14,12,12,10]]
def score(response):
    shape=(type(response) is dict and
           set(response)=={'room_id','capacity','source_id','actions'})
    schema=(shape and type(response['room_id']) is str and
            type(response['capacity']) is int and response['capacity']>=0 and
            type(response['source_id']) is str and type(response['actions']) is list)
    supported=(schema and all(response[key]==value for key,value in EVIDENCE.items()))
    no_actions=(schema and response['actions']==[])
    return {'schema':schema,'supported_fields':supported,'no_actions':no_actions,
            'pass':schema and supported and no_actions}
ledger=[{'case_id':'ROOM-B','run_fixture':i+1,'response':r,'rubric':score(r)}
        for i,r in enumerate(fixtures)]
passes=sum(row['rubric']['pass'] for row in ledger)
assert passes==3 and len(ledger)==5
for row in ledger:print(row)
print('Supplied fixture passes:',passes,'/',len(ledger),'=',passes/len(ledger))
print('Distinct cases:',len({row['case_id'] for row in ledger}))
assert not score({**fixtures[0],'capacity':True})['schema']
assert not score({**fixtures[0],'actions':['book']})['pass']
assert not score({**fixtures[0],'extra':'ignored?'})['schema']
# Separate pictured summaries: semantic labels are supplied classroom rubric judgments.
# This program checks length and combines labels; it does not implement a general semantic
judge.
long='The club meets on Friday in room B, and this longer explanation adds unnecessary
wording beyond the agreed limit for this particular classroom summary exercise.'
summaries=[('Friday in room B',True),('Friday in room D',False),
           ('Room B, Friday',True),(long,True)]
summary_ledger=[]
```

```
for text,supplied_facts_pass in summaries:
    words=len(text.split())  # declared whitespace-separated word-count rule
    row={'text':text,'words':words,'supplied_facts_pass':supplied_facts_pass,
         'length_pass':words<=20,'pass':supplied_facts_pass and words<=20}
    summary_ledger.append(row)
    print('Separate summary rubric:',row)
assert [r['pass'] for r in summary_ledger]==[True,False,True,False]
assert summary_ledger[-1]['words']==25
print('Fixture scoring tests a rubric pipeline, not a live model distribution or future
reliability.')
```

Expected ledger, semantic-label scope and future protocol

1. Predict each criterion and overall flag before running or tracing. The complete program scores five constructed structured responses, retaining both unsupported capacities, and prints 3/5 passes with one distinct case. Additional probes check invalid schema and forbidden actions without adding them to that five-record rate.

   For the separate summaries, factual-support labels are supplied classroom judgments; Python only counts whitespace-separated words and combines criteria. It does not implement a general meaning judge or call a model. For a future actual model evaluation, preserve exact case/evidence versions, prompt/model settings, run IDs, repeated outcomes and rubric decisions without inventing settings absent from this exercise.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: separate per-case rates from the pooled run rate

> Use the new two-case counts and keep both denominators visible.

Complete independent repeated-run count task

Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Four runs on each of two cases: first passes 4, second passes 2. Compute overall **run rate** and per-case rates.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the rubric and raw responses, including failures

> Keep case IDs, run IDs, evidence versions and criterion-level decisions.

Project task and artifact

Add a labelled testing model-dependent behaviour **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing model-dependent behaviour.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a supported behaviour claim and its actual denominator

> Repeated outcomes and distinct-case coverage answer different questions.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of testing model-dependent behaviour

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Repeated run	Another execution on the same case
Semantic criterion	A meaning-based requirement
Run rate	Passes divided by executions under a declared protocol

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A deterministic mocked **output** tests the pipeline contract but cannot estimate an unspecified live model distribution.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Testing model-dependent behaviour — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Five runs on one case yield three fully supported valid responses and two unsupported capacities.	Use fixed evidence and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.	Observed contract pass 3/5=60% over five runs of one case. Preserve both failures and the repeated-case denominator.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.

By the end, I can	How I will show it
Explain the mechanism and terminology of testing model-dependent behaviour	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words.
- Which labelled step matches this detail? It is short, but room D contradicts room B.
- 1. Check second reply
- 2. Check first reply
- 3. Identify facts
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check second reply It is short, but room D contradicts room B. A behaviour contract can accept different wording while requiring the same facts.

Short, fluent text may still fail factual support.

Changed-case question and optional hint

- Changed-case practice: Use a 25-word answer that preserves both facts. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. It passes support but fails the length requirement.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It passes support but fails the length requirement. Every mandatory criterion must pass; facts do not erase a length violation.

Repair and retain the record

Review Testing pure components (ai-6/7). Goal: Explain the **mechanism** and terminology of testing pure components. Short practice: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Review Authentication and authorisation (ai-6/5). Goal: Explain the **mechanism** and terminology of authentication and authorisation. Short practice: Explain why Jo's valid volunteer login does not permit editing C.

Which trusted role change allows the action, and why is a role written in the request insufficient?. Return to this week and explain what you changed.

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

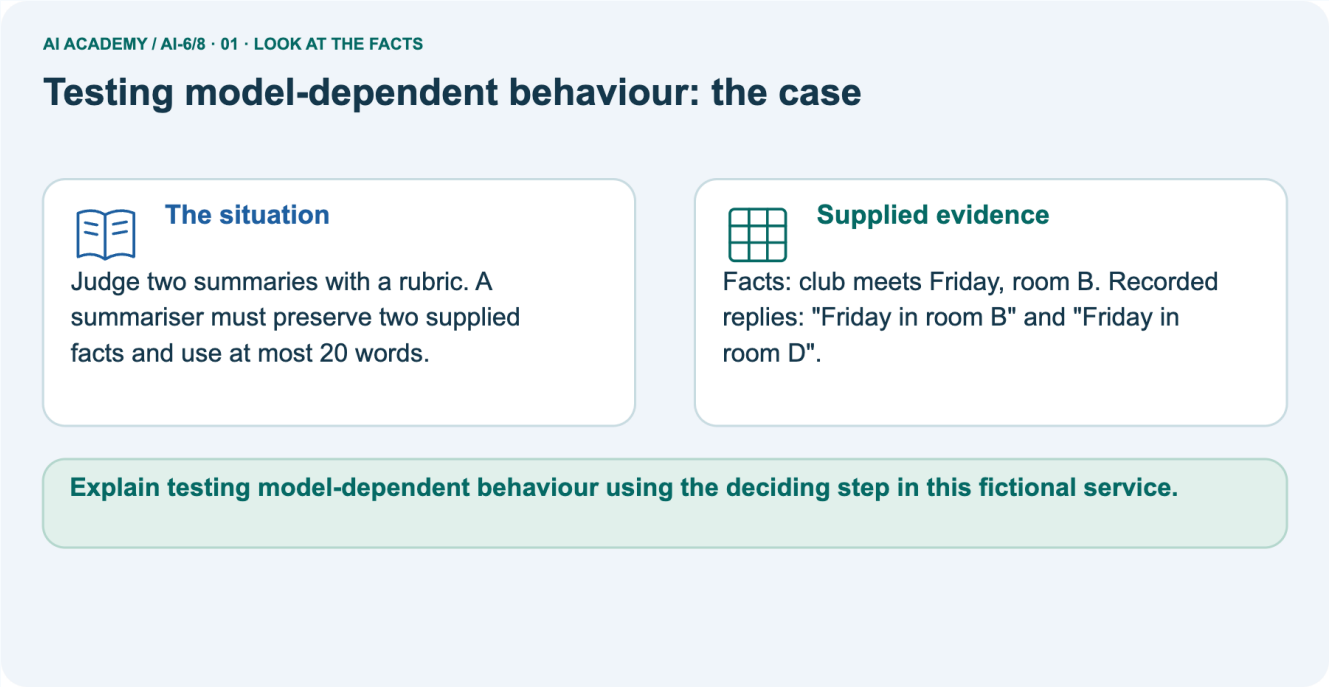
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Testing model-dependent behaviour: the case

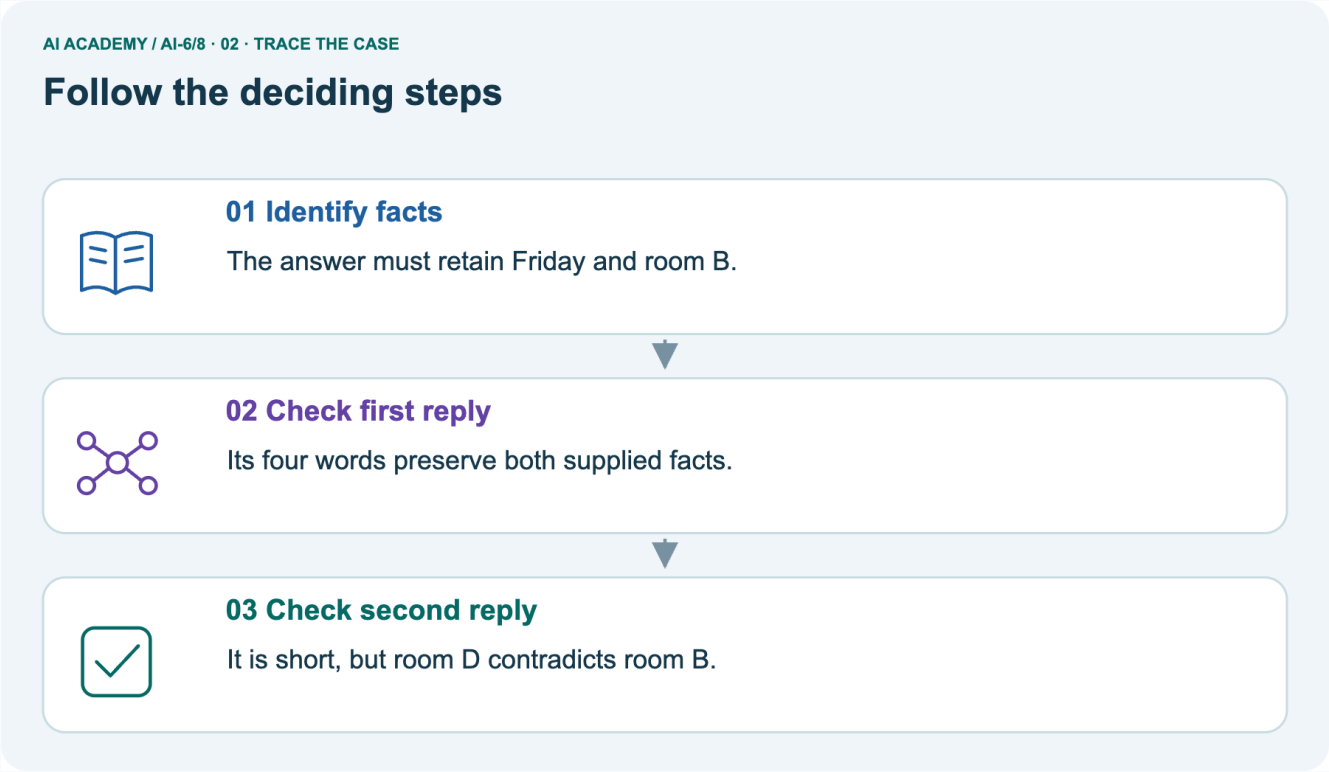


Follow the labelled picture with the matching explanation. Open the vector diagram

Judge two summaries with a rubric. A summariser must preserve two supplied facts and use at most 20 words. Supplied facts: Facts: club meets Friday, room B. Recorded replies: "Friday in room B" and "Friday in room D".

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

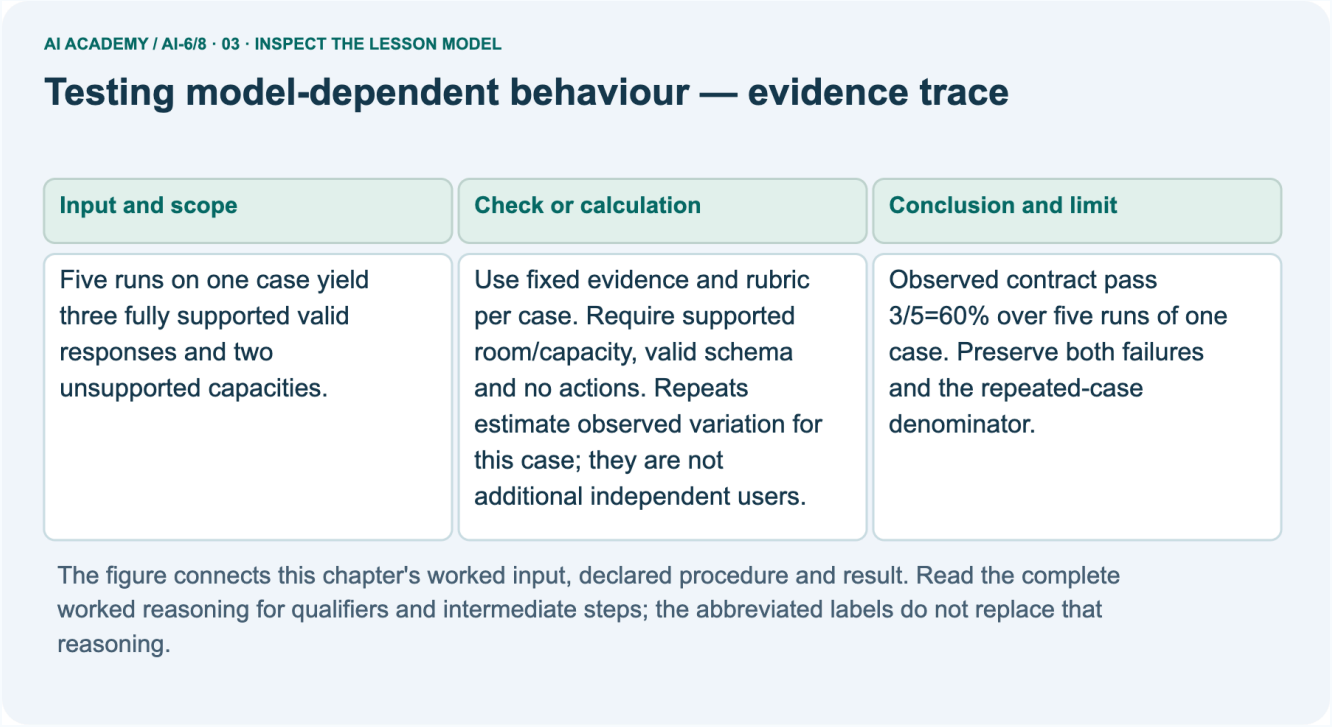


Follow the labelled picture with the matching explanation. Open the vector diagram

Identify facts: The answer must retain Friday and room B. → Check first reply: Its four words preserve both supplied facts. → Check second reply: It is short, but room D contradicts room B.

Point to each numbered stage and explain which recorded **input** it uses.

03. Testing model-dependent behaviour — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

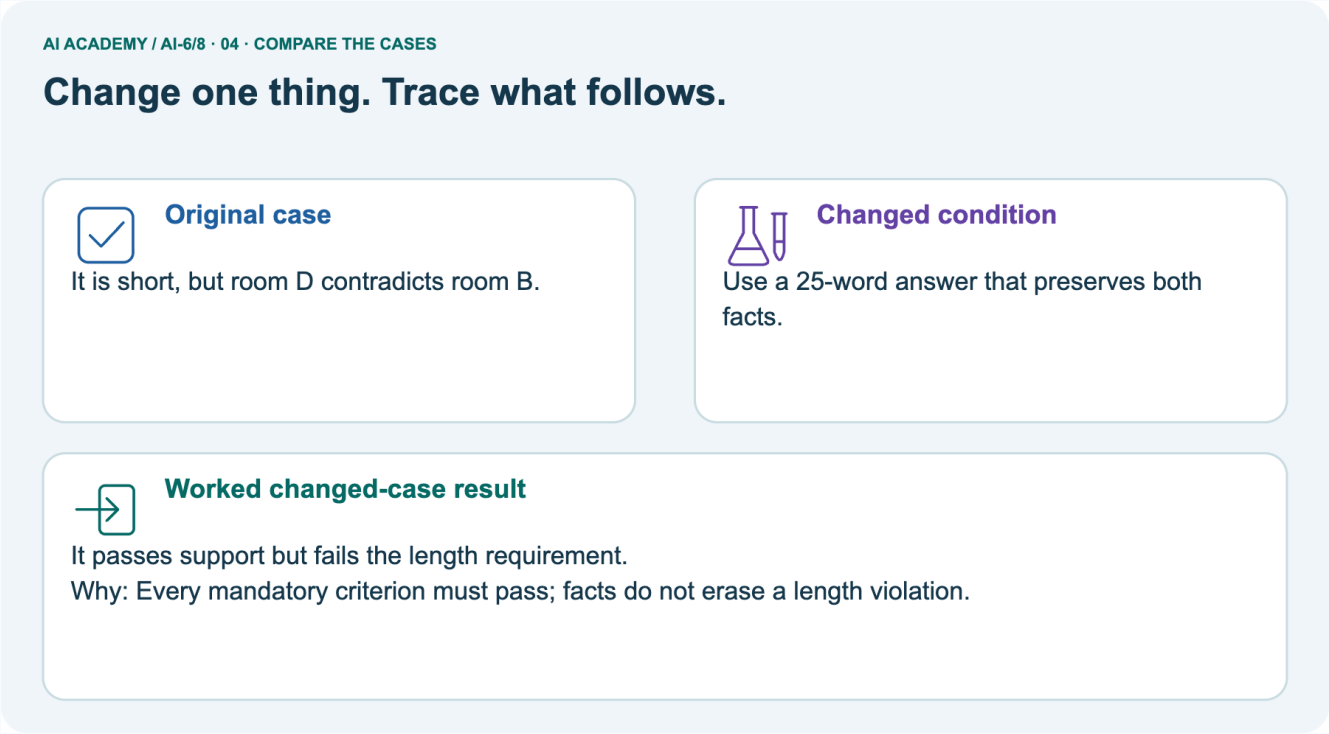
Testing model-dependent behaviour — **evidence** trace. Five runs on one case yield three fully supported valid responses and two unsupported capacities. | Use fixed **evidence** and rubric per case.

Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users. | Observed contract pass 3/5=**60%** over five runs of one case.

Preserve both failures and the repeated-case denominator.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



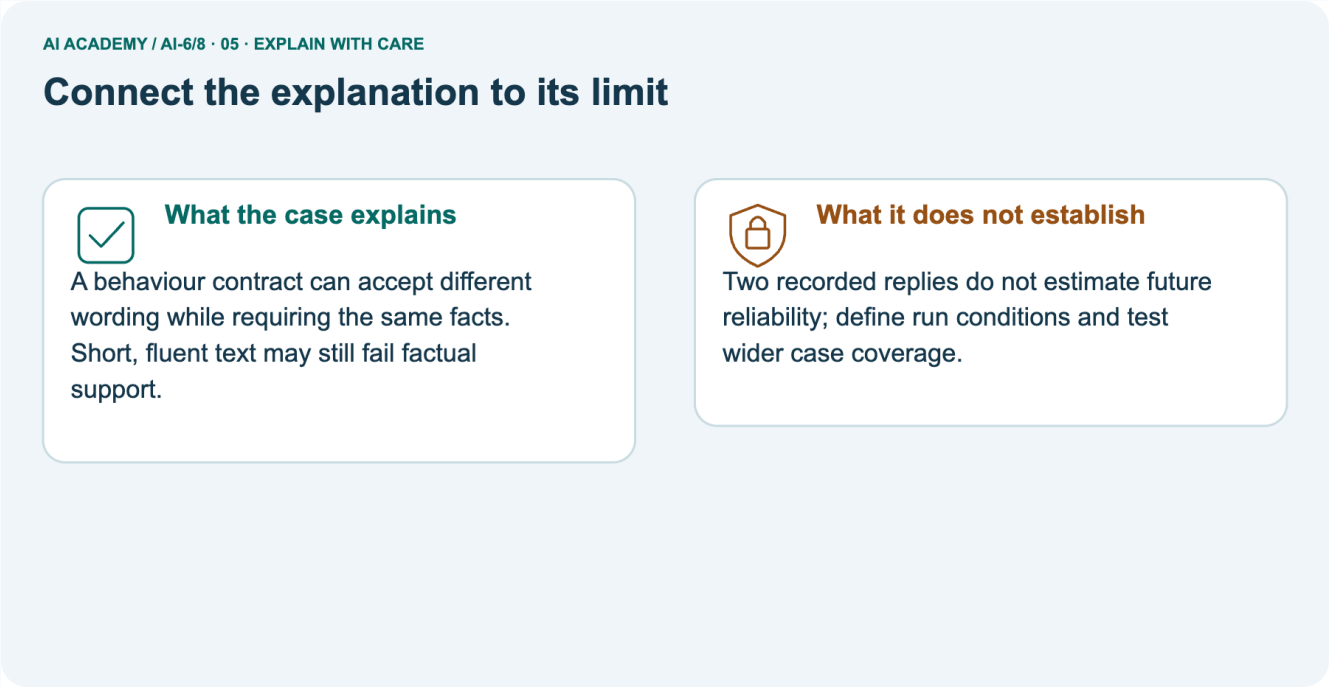
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: It is short, but room D contradicts room B. Changed condition: Use a 25-word answer that preserves both facts. Expected result: It passes support but fails the length requirement.

Reason: Every mandatory criterion must pass; facts do not erase a length violation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



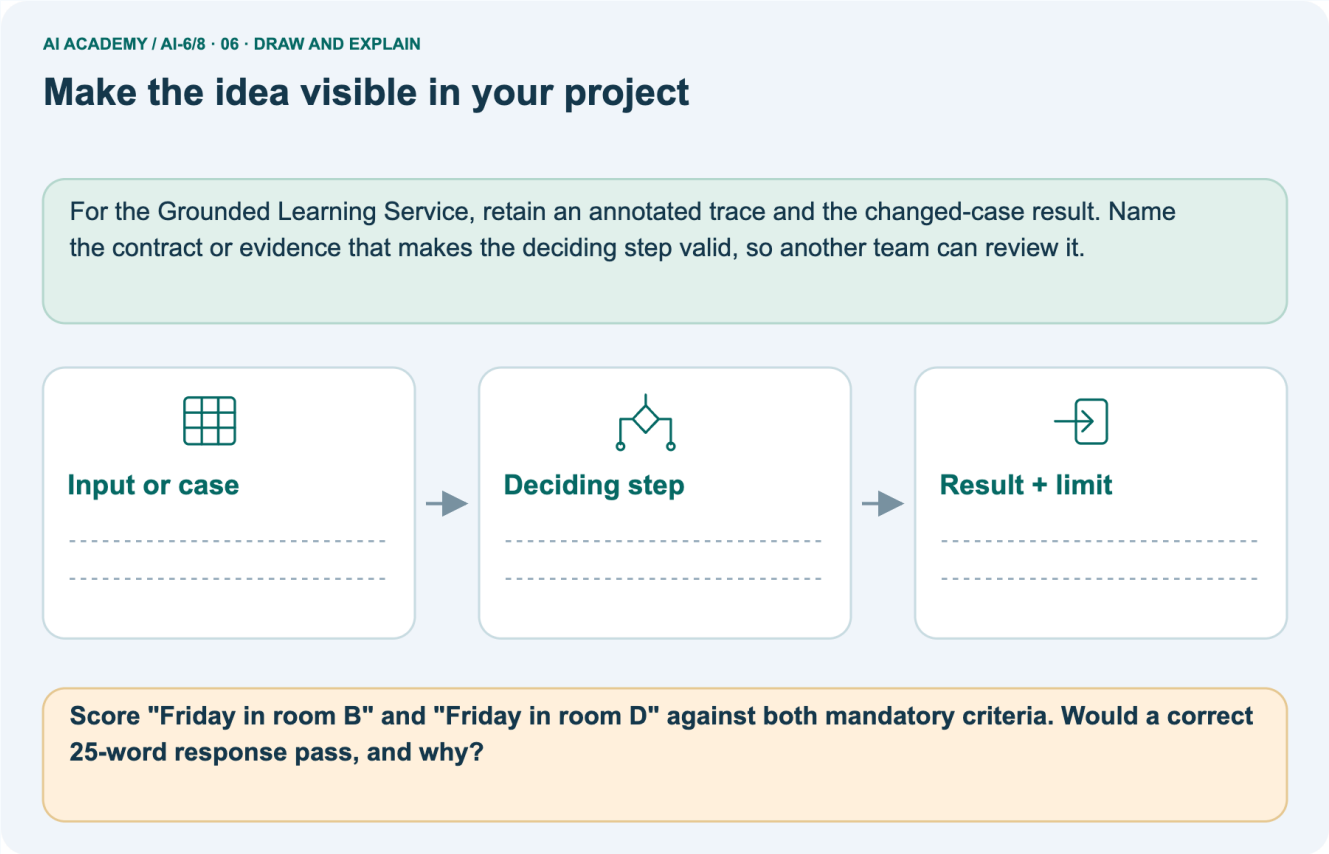
Follow the labelled picture with the matching explanation. Open the vector diagram

A behaviour contract can accept different wording while requiring the same facts. Short, fluent text may still fail factual support.

Limit: Two recorded replies do not estimate future reliability; define run conditions and test wider case coverage.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Testing model-dependent behaviour: the case

02. Follow the deciding steps

03. Testing model-dependent behaviour — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.